BJT amplifiers

Input resistance, output resistance, and voltage gain

24 progressive exercises with diagrams and worked solutions

Built around the BJT amplifier topics in your Lectures 5–8 and Sheets 6–7. All exercises are new. Start with the first three pages, then solve the questions before opening the solutions.

The five-step routine

1. **Find the DC bias first, if it is not given.** Capacitors are open for DC. Compute I_C and verify the region of operation. Then use $g_m = I_C/V_T$ and $r_\pi = \beta/g_m$.
2. **Redraw for midband AC.** Ground ideal DC voltage supplies; open ideal DC current sources. Short sufficiently large coupling/bypass capacitors. Keep every resistor that still connects two different AC nodes.
3. **Name the ports.** Here v_i means the actual voltage at the amplifier input terminal. The source voltage v_s is usually different because R_s and R_{in} form a divider.
4. **Find loaded voltage gain.** Include R_L, or the next stage's input resistance, in the output load. Find $A_v = v_o/v_i$ first, then $A_s = v_o/v_s = A_v R_{\text{in}}/(R_s + R_{\text{in}})$ when the input divider applies.
5. **Find output resistance independently.** Remove the external load, set independent signal sources to zero, retain their internal resistances, and apply a test voltage v_x. Keep the controlled source: $R_{\text{out}} = v_x/i_x$.

What setting a source to zero means: an ideal voltage source becomes a short; an ideal current source becomes an open. A voltage source with series resistance leaves that resistance in the circuit.

A port checklist

Quantity	What you must include
R_{in}	Everything connected inside the input port, including bias resistors. Exclude the source's own series resistance.
$A_v = v_o/v_i$	The actual output load. Specify whether v_i is at base or emitter.
$A_s = v_o/v_s$	Source attenuation as well as loaded stage gain.
R_{out}	Source resistance after zeroing its voltage. Remove only the external load, not an internal emitter or collector resistor.

Conventions throughout. $V_T = 25$ mV; transistors are forward-active; $r_o = \infty$ unless stated. R_B means the equivalent AC bias resistance, usually $R_1 \parallel R_2$. Figures are midband AC circuits with bias sources omitted, except the explicitly labeled DC exercise; bypass capacitors are shown where their connections are being tested. All numerical gains are V/V. Use finite β unless an approximation is requested.

Suggested route. First solve 1–6, then 9–14, then 7–8 and 15–24. Allow about 10–15 minutes per exercise, longer for cascades. Solutions begin at page 16.

A compact map of the formulas

Use these only after identifying the topology and the port. Let $R'_C = R_C \parallel R_L$ and $R'_E = R_E \parallel R_L$ where relevant. These expressions assume $r_o = \infty$.

Topology	Input resistance of transistor path	Loaded stage gain
CE, grounded emitter	r_π	$-g_m R'_C$
CE, unbypassed R_E	$r_\pi + (\beta+1)R_E$	$-\beta R'_C/[r_\pi + (\beta+1)R_E]$
Emitter follower	$r_\pi + (\beta+1)R'_E$	$(\beta+1)R'_E/[r_\pi + (\beta+1)R'_E]$
CB, grounded base	$r_\pi/(\beta+1)$	$+g_m R'_C$
CB, base grounded through R_B	$(r_\pi + R_B)/(\beta+1)$	$+\beta R'_C/(r_\pi + R_B)$

For CE and follower input ports, put the bias resistance R_B in parallel with the transistor path. For CB, any external emitter-to-ground resistor is in parallel with the transistor input resistance.

For CE and CB with $r_o = \infty$, $R_{\text{out}} = R_C$. For a follower,

$$R_{\text{out}} = R_E \parallel \frac{r_\pi + (R_s \parallel R_B)}{\beta + 1},$$

where the external R_L has been removed. If no base-bias resistor exists, use R_s in place of $R_s \parallel R_B$.

Worked warm-up: the three gains people mix up

A grounded-emitter CE stage has $I_C = 1$ mA, $\beta = 100$, $R_C = 2\,\text{k}\Omega$, $R_L = 2\,\text{k}\Omega$, $R_B = 10\,\text{k}\Omega$, and $R_s = 500\,\Omega$.

$$g_m = 40\,\text{mS}, \quad r_\pi = 2.5\,\text{k}\Omega, \quad R_{\text{in}} = 10\,\text{k}\Omega \parallel 2.5\,\text{k}\Omega = 2\,\text{k}\Omega.$$

$$A_{v,\text{unloaded}} = -g_m R_C = -80, \quad A_{v,\text{loaded}} = -g_m(R_C \parallel R_L) = -40.$$

$$\boxed{A_s = -40\frac{2000}{500 + 2000} = -32, \qquad R_{\text{out}} = 2\,\text{k}\Omega.}$$

Interpretation: the collector load halves the stage gain, and the source divider reduces it again. Output resistance is not $R_C \parallel R_L$ because R_L is removed for that measurement.

Two useful physical checks

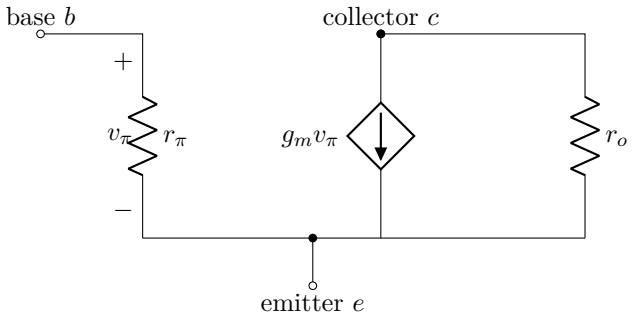
Looking into the base: emitter resistance is multiplied by $\beta+1$ because emitter current is $\beta+1$ times base current.

Looking into the emitter: base-side resistance is divided by $\beta + 1$. This is why a follower has low output resistance and a CB stage has low input resistance.

Source context: Lectures 5–8 and Sheets 6–7. The source material guided the topics; the problems and solutions here were independently constructed.

Why the formulas work: replace the transistor

The low-frequency hybrid-π model below keeps the physical base, collector and emitter terminals. $v_\pi = v_b - v_e$ controls the current source; it is not always equal to the applied input voltage.



The diamond current source points from collector to emitter. If Early effect is neglected, remove the r_o branch. Do not remove the controlled source when finding output resistance: its controlling voltage must be determined from the zeroed-source circuit.

Derive one CE result instead of memorizing it

Let an unbypassed resistor R_E connect emitter to ground and assume $r_o = \infty$. Apply the input at the base. Then

$$i_b = \frac{v_\pi}{r_\pi}, \quad i_c = g_m v_\pi = \beta i_b, \quad i_e = (\beta + 1) i_b.$$

The input voltage must cover both the base–emitter voltage and the emitter-resistor drop:

$$v_i = v_\pi + v_e = i_b r_\pi + (\beta + 1) i_b R_E.$$

So the resistance of the transistor path, seen from the base, is

$$R_{ib} = \frac{v_i}{i_b} = r_\pi + (\beta + 1) R_E.$$

The output at the collector is $v_o = -i_c (R_C \parallel R_L)$. Dividing by the input expression gives

$$A_v = -\frac{\beta (R_C \parallel R_L)}{r_\pi + (\beta + 1) R_E}.$$

Only after this do you put R_B in parallel with R_{ib} to get the whole amplifier's input resistance. R_s belongs outside that input port.

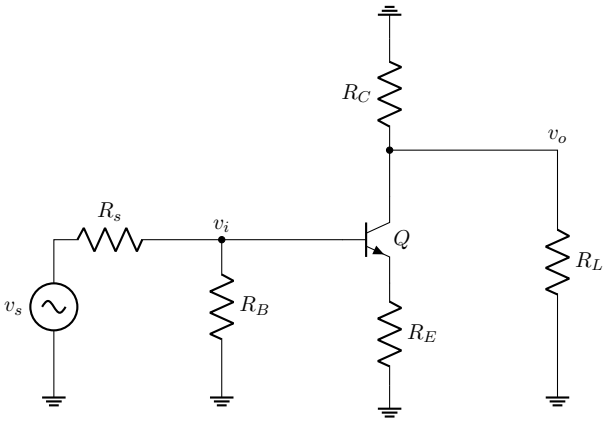
A follower output-resistance measurement

Remove the external load and set $v_s = 0$. The base is connected to ground through $R_G = R_s \parallel R_B$, not necessarily through a short. A test current entering the emitter is shared between the internal R_E and the transistor path. With $r_o = \infty$, that transistor path has resistance

$$R_{e,\text{path}} = \frac{r_\pi + R_G}{\beta + 1}, \qquad R_{\text{out}} = R_E \parallel R_{e,\text{path}}.$$

Ask before every calculation: Which terminal is the input? Which terminal is the output? Which terminal is at AC ground? Which resistor is the external load?

1. Grounded-emitter foundations



CE stage. $R_E = 0$ means a wire; $R_B = \infty$ or $R_L = \infty$ means omit that branch.

Exercise 1 — Start with a grounded emitter

$I_C = 1$ mA, $\beta = 100$, $R_C = 2000\,\Omega$, $R_E = 0$, $R_s = 0$, $R_B = R_L = \infty$. Find g_m, r_π, R_{in}, A_v and R_{out}. State the phase relationship. For $v_i = 1$ mV peak, find the output peak amplitude.

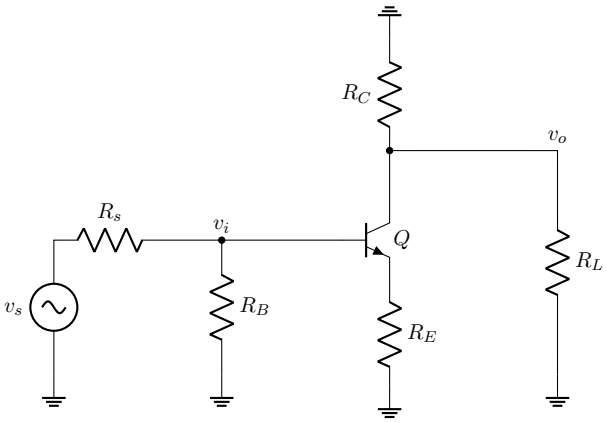
Worked solution

Exercise 2 — Halve the current, double the resistor

$I_C = 0.5$ mA, $\beta = 100$, $R_C = 4000\,\Omega$, $R_E = 0$, $R_s = 0$, $R_B = R_L = \infty$. Find g_m, r_π, R_{in}, A_v and R_{out}. State the phase relationship. Compare your answers with Exercise 1: which quantities stay the same?

Worked solution

2. Loading and Early effect



CE stage. $R_E = 0$ means a wire; $R_B = \infty$ or $R_L = \infty$ means omit that branch.

Exercise 3 — The source is not the base

$I_C = 1$ mA, $\beta = 100$, $R_C = 3\,\text{k}\Omega$, $R_L = 6\,\text{k}\Omega$, $R_B = 15\,\text{k}\Omega$, $R_s = 1\,\text{k}\Omega$, $R_E = 0$. Find $g_m, r_\pi, R_{\text{in}}, A_v, A_s, R_{\text{out}}$. For $v_s = 2$ mV peak, find v_i and the output peak amplitude.

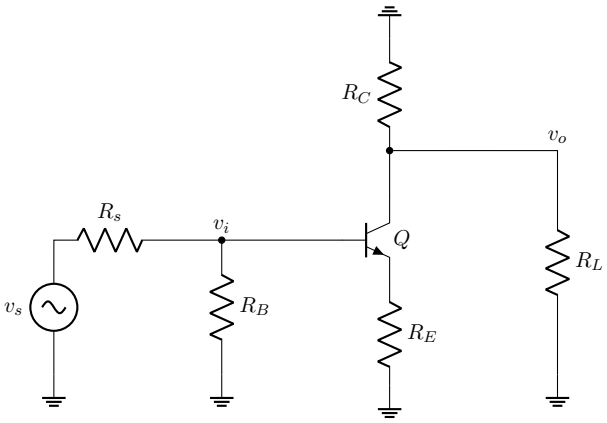
Worked solution

Exercise 4 — Early effect with a grounded emitter

Repeat Exercise 3 with $V_A = 20$ V. Use $r_o \simeq V_A/I_C$; retain the same bias current. Find $R_{\text{in}}, A_v, A_s, R_{\text{out}}$ and explain why r_o changes the collector-side results but not the input resistance here.

Worked solution

3. Emitter degeneration



CE stage. $R_E = 0$ means a wire; $R_B = \infty$ or $R_L = \infty$ means omit that branch.

Exercise 5 — Add emitter degeneration

$I_C = 1$ mA, $\beta = 100$, $R_C = 3\,\text{k}\Omega$, $R_L = 6\,\text{k}\Omega$, $R_B = 15\,\text{k}\Omega$, $R_s = 1\,\text{k}\Omega$ and $R_E = 100\,\Omega$. Derive R_{ib} and A_v from $v_i = i_b r_\pi + i_e R_E$. Find $R_{\text{in}}, A_s, R_{\text{out}}$. Compare A_v with $-g_m(R_C \parallel R_L)/(1 + g_m R_E)$.

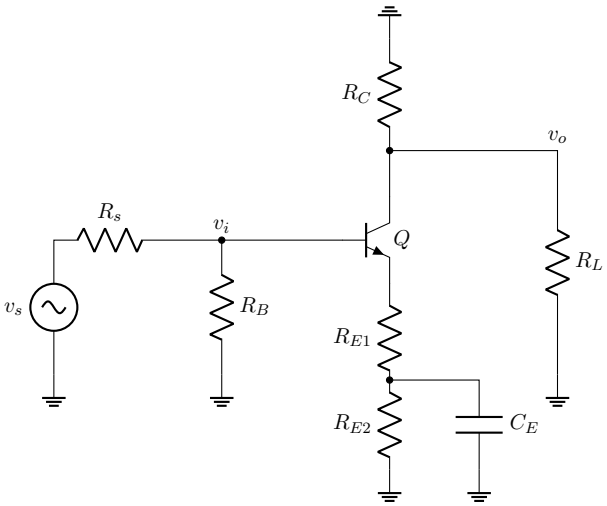
Worked solution

Exercise 6 — Do not silently replace beta plus one

$I_C = 1$ mA, $\beta = 50$, $R_C = 3\,\text{k}\Omega$, $R_L = 6\,\text{k}\Omega$, $R_B = 15\,\text{k}\Omega$, $R_s = 1\,\text{k}\Omega$ and $R_E = 470\,\Omega$. Derive R_{ib} and A_v from $v_i = i_b r_\pi + i_e R_E$. Find $R_{\text{in}}, A_s, R_{\text{out}}$. Compare A_v with $-g_m(R_C \parallel R_L)/(1 + g_m R_E)$.

Worked solution

4. Bypass capacitors



CE stage. $R_E = 0$ means a wire; $R_B = \infty$ or $R_L = \infty$ means omit that branch.

Exercise 7 — Only part of the emitter resistor is bypassed

For the split-emitter figure, $I_C = 1$ mA, $\beta = 100$, $R_C = R_L = 4\,\text{k}\Omega$, $R_B = 20\,\text{k}\Omega$, $R_s = 500\,\Omega$, $R_{E1} = 100\,\Omega$ and $R_{E2} = 900\,\Omega$. C_E is a midband short. Find the emitter resistance used for AC and for DC. Then calculate $R_{\text{in}}, A_v, A_s, R_{\text{out}}$.

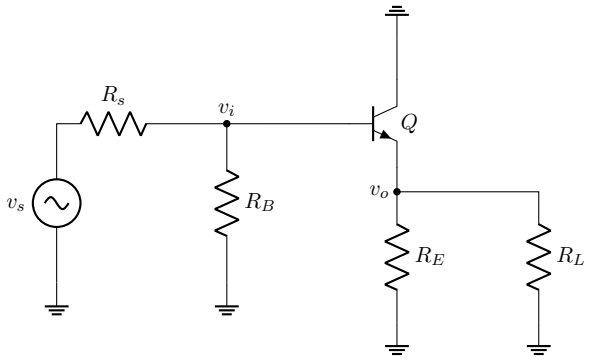
Worked solution

Exercise 8 — What changes when the bypass capacitor is removed?

Use Exercise 7 but physically remove C_E. Recalculate $R_{\text{in}}, A_v, A_s, R_{\text{out}}$. Does the DC operating point change? Explain why using $R_E = 0$ would be wrong both before and after this removal.

Worked solution

5. Emitter followers



Emitter follower. The collector is at AC ground, not physically shorted to ground in DC.

Exercise 9 — The basic emitter follower

$I_C = 1$ mA, $\beta = 100$, $R_E = R_L = 1\,\text{k}\Omega$, $R_B = \infty$, $R_s = 0$. Find the transistor input resistance, loaded base-to-emitter gain, and output resistance with the external load removed. Explain why a gain below one does not make this stage useless.

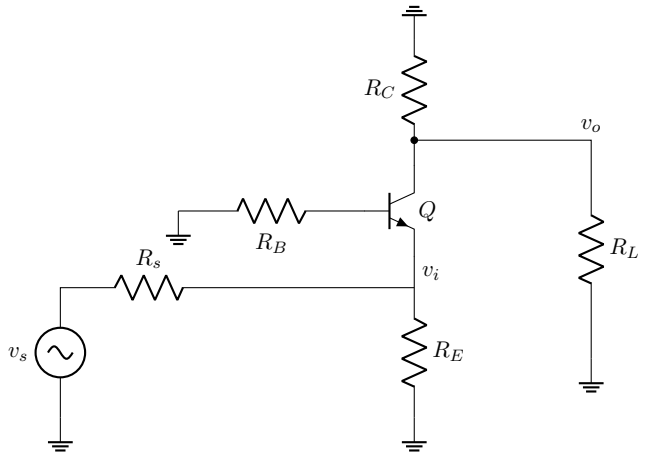
Worked solution

Exercise 10 — Source resistance comes back during Rout

Use the same follower with $R_B = 20\,\text{k}\Omega$ and $R_s = 3\,\text{k}\Omega$. Find $R_{\text{in}}, A_v, A_s, R_{\text{out}}$. A student claims $R_{\text{out}} = 1/g_m = 25\,\Omega$. Explain what this estimate ignores.

Worked solution

6. Common-base foundations



Common-base stage. $R_B = 0$ shorts the base to AC ground; $R_E = \infty$ omits the emitter resistor.

Exercise 11 — Common base: input at the emitter

$I_C = 1$ mA, $\beta = 100$, $R_C = 2\,\text{k}\Omega$, $R_L = \infty$, $R_E = \infty$, $R_B = 0$, $R_s = 0$. Derive the emitter input resistance and voltage gain. Explain why a positive emitter-voltage change raises the collector voltage.

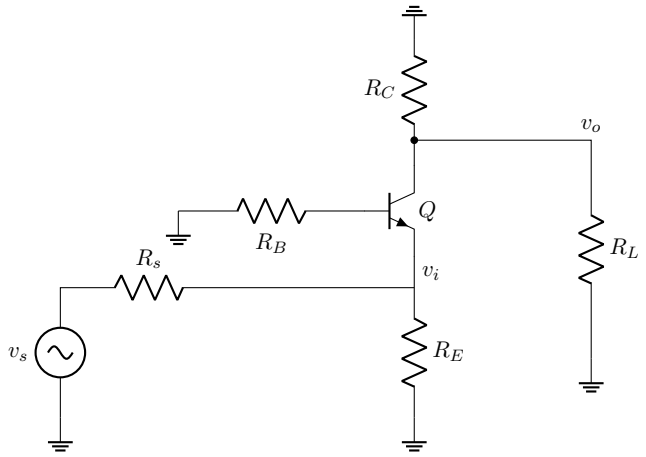
Worked solution

Exercise 12 — A high stage gain can hide a small overall gain

$I_C = 1$ mA, $\beta = 100$, $R_C = R_L = 2\,\text{k}\Omega$, $R_E = 1\,\text{k}\Omega$, $R_B = 0$, $R_s = 100\,\Omega$. Find $R_{\text{in}}, A_v, A_s, R_{\text{out}}$. Explain why A_s is much smaller than A_v.

Worked solution

7. Finite base resistance



Common-base stage. $R_B = 0$ shorts the base to AC ground; $R_E = \infty$ omits the emitter resistor.

Exercise 13 — The base is grounded through a resistor

$I_C = 1$ mA, $\beta = 100$, $R_C = R_L = 2\,\text{k}\Omega$, $R_E = \infty$, $R_B = 2.5\,\text{k}\Omega$, $R_s = 100\,\Omega$. Derive v_π / v_i, then find $R_{\text{in}}, A_v, A_s, R_{\text{out}}$. Why is it wrong to set $v_b = 0$?

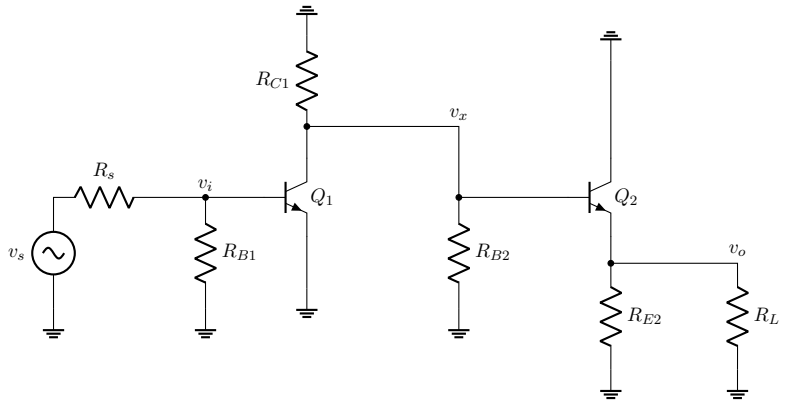
Worked solution

Exercise 14 — Compare two base terminations

Use $I_C = 1$ mA, $\beta = 100$, $R_C = R_L = 2\,\text{k}\Omega$, $R_E = 1\,\text{k}\Omega$, $R_s = 100\,\Omega$. Calculate R_{in}, A_v, A_s for (i) $R_B = 0$ and (ii) $R_B = 10\,\text{k}\Omega$. Decide whether the increased emitter input resistance necessarily improves the overall voltage gain.

Worked solution

8. CE–follower cascades



Midband cascade: interstage coupling capacitor is replaced by a short. Biases are set independently.

Exercise 15 — CE driving a follower

For the CE–follower cascade, $I_{C1} = 1$ mA, $I_{C2} = 2$ mA, $\beta_1 = \beta_2 = 100$, $R_{B1} = 20\,\text{k}\Omega$, $R_{B2} = 50\,\text{k}\Omega$, $R_{C1} = 3\,\text{k}\Omega$, $R_{E2} = 1\,\text{k}\Omega$, $R_L = 500\,\Omega$, and $R_s = 1\,\text{k}\Omega$. Find $R_{\text{in1}}, R_{\text{in2}}, A_1 = v_x/v_i, A_2 = v_o/v_x, A_v, A_s$, and the complete cascade's output resistance.

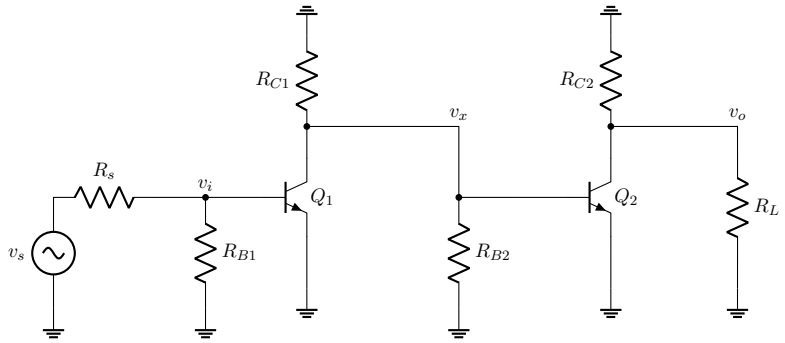
Worked solution

Exercise 16 — Change the final load and follow the effect backward

Repeat Exercise 15 with $R_L = 50\,\Omega$. Explain why the first-stage gain also changes, even though R_{C1} and I_{C1} did not change. Does the measured output resistance change when the external load value changes?

Worked solution

9. CE–CE cascades



Midband cascade: interstage coupling capacitor is replaced by a short. Biases are set independently.

Exercise 17 — Two CE stages: do not multiply unloaded gains

In the CE–CE cascade, $I_{C1} = I_{C2} = 1$ mA, $\beta_1 = \beta_2 = 100$, $R_{B1} = 20\,\text{k}\Omega$, $R_{B2} = 10\,\text{k}\Omega$, $R_{C1} = 3\,\text{k}\Omega$, $R_{C2} = R_L = 2\,\text{k}\Omega$, $R_s = 500\,\Omega$. Find both input resistances, the loaded gain of each stage, A_v, A_s and output resistance. Compare A_v with the product of both stages' unloaded gains.

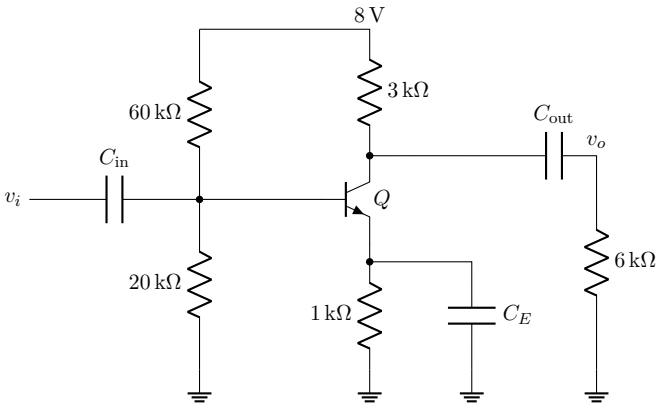
Worked solution

Exercise 18 — An extra interstage resistor

Use Exercise 17 but insert $R_x = 1\,\text{k}\Omega$ in the interstage wire, between the first collector (v_x) and the second base (v_{b2}). Find v_x/v_i, v_{b2}/v_x, v_o/v_{b2}, A_v and A_s. Is the first-stage gain alone sufficient to judge the overall improvement?

Worked solution

10. DC bias to AC performance



DC circuit for Exercises 19–20. All capacitors are large for midband AC.

Exercise 19 — From the DC circuit to the AC circuit

Use the DC figure with $\beta = 100$, $V_{BE} = 0.70$ V, $V_A = \infty$. The AC source driving C_{in} is ideal. Calculate $I_C, V_B, V_E, V_C, V_{CE}$ and check forward-active operation. Draw the midband model and calculate $g_m, r_\pi, R_{\text{in}}, A_v, R_{\text{out}}$ with C_E connected.

Worked solution

Exercise 20 — Remove the bypass and add a realistic source

In Exercise 19, remove C_E and drive C_{in} from a source with $R_s = 1\,\text{k}\Omega$. Calculate the new $R_{\text{in}}, A_v, A_s, R_{\text{out}}$. Explain why neither removing C_E nor changing the AC source resistance changes the DC bias in this circuit.

Worked solution

11. Finite-output-resistance challenges

Use the CE circuit from page 6 for Exercise 21 and the follower circuit from page 8 for Exercise 22. Those figures retain their labeled topology; the values below replace earlier values.

Focus: a finite r_o connects collector to emitter. It is a resistor to ground only when the opposite terminal is actually at AC ground.

Exercise 21 — Challenge: Rout with degeneration and finite ro

Use the CE figure with $I_C = 1$ mA, $\beta = 100$, $V_A = 50$ V, $R_C = 4\,\text{k}\Omega$, $R_E = 200\,\Omega$, $R_B = \infty$, and $R_s = 0$. Remove R_L. Find r_o and derive the resistance looking into the transistor collector using a test voltage, then include R_C. Compare with the incorrect shortcut $R_C \parallel r_o$. Do not assume the emitter is at AC ground.

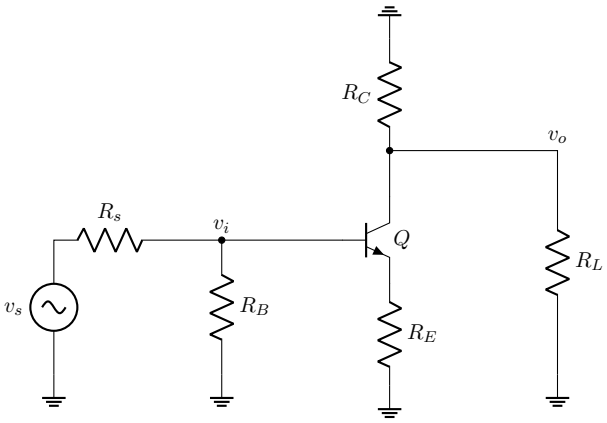
Worked solution

Exercise 22 — Challenge: finite ro in a follower

Use the emitter-follower figure with $I_C = 1$ mA, $\beta = 100$, $r_o = 2\,\text{k}\Omega$ (given directly), $R_E = R_L = 1\,\text{k}\Omega$, $R_B = 20\,\text{k}\Omega$, $R_s = 3\,\text{k}\Omega$. Find $R_{\text{in}}, A_v, A_s, R_{\text{out}}$. Explain where r_o appears in the AC circuit.

Worked solution

12. Design and error detection



CE stage. $R_E = 0$ means a wire; $R_B = \infty$ or $R_L = \infty$ means omit that branch.

Exercise 23 — Design for a specified loaded gain

Use the CE figure with $I_C = 1$ mA, $\beta = 100$, $R_C = R_L = 4\,\text{k}\Omega$, $R_B = 50\,\text{k}\Omega$, $R_s = 1\,\text{k}\Omega$, and $r_o = \infty$. Select the unbypassed R_E so the *base-to-output loaded gain* is exactly -10. Then find $R_{\text{in}}, A_s, R_{\text{out}}$. Explain why designing $A_v = -10$ does not make $A_s = -10$. The bias circuit is adjusted to preserve I_C.

Worked solution

Exercise 24 — Diagnose a complete wrong solution

Use $I_C = 1$ mA, $\beta = 100$, $R_C = 3\,\text{k}\Omega$, $R_L = 6\,\text{k}\Omega$, $R_B = 15\,\text{k}\Omega$, $R_s = 1\,\text{k}\Omega$, $R_E = 100\,\Omega$, $r_o = \infty$. A student writes:

$$R_{\text{in}} = R_s + r_\pi, \quad A_v = -g_m R_C, \quad A_s = A_v, \quad R_{\text{out}} = R_C \parallel R_L.$$

Identify every conceptual mistake, derive the corrected quantities, and calculate the output amplitude for $v_s = 1$ mV peak.

Worked solution

Worked solutions

Use these to check the circuit reasoning, not only the final numbers. Rounding to three significant figures is sufficient.

Solution 1 — Start with a grounded emitter

First $g_m = 40\,\text{mS}$ and $r_\pi = 2500\,\Omega$. The transistor input path is

$$R_{ib} = r_\pi + (\beta + 1)R_E = 2500\,\Omega, \quad R_{\text{in}} = R_B \parallel R_{ib} = 2500\,\Omega.$$

The effective collector load is $R_C \parallel R_L = 2000\,\Omega$. Hence

$$A_v = -\frac{\beta(2000)}{2500 + (\beta + 1)(0)} = -80, \qquad A_s = A_v \frac{2500}{0 + 2500} = -80.$$

With the load removed and the source zeroed, $R_{\text{out}} = 2000\,\Omega$. The output is inverted and its peak amplitude is 80 mV.

Solution 2 — Halve the current, double the resistor

First $g_m = 20\,\text{mS}$ and $r_\pi = 5000\,\Omega$. The transistor input path is

$$R_{ib} = r_\pi + (\beta + 1)R_E = 5000\,\Omega, \quad R_{\text{in}} = R_B \parallel R_{ib} = 5000\,\Omega.$$

The effective collector load is $R_C \parallel R_L = 4000\,\Omega$. Hence

$$A_v = -\frac{\beta(4000)}{5000 + (\beta + 1)(0)} = -80, \qquad A_s = A_v \frac{5000}{0 + 5000} = -80.$$

With the load removed and the source zeroed, $R_{\text{out}} = 4000\,\Omega$. The gain stays -80; $r_p i$, input resistance and output resistance double relative to Exercise 1. The output is inverted.

Worked solutions: 3–4

Solution 3 — The source is not the base

First $g_m = 40\,\text{mS}$ and $r_\pi = 2500\,\Omega$. The transistor input path is

$$R_{ib} = r_\pi + (\beta+1)R_E = 2500\,\Omega, \quad R_{\text{in}} = R_B \parallel R_{ib} = 2143\,\Omega.$$

The effective collector load is $R_C \parallel R_L = 2000\,\Omega$. Hence

$$A_v = -\frac{\beta(2000)}{2500 + (\beta+1)(0)} = -80, \qquad A_s = A_v \frac{2143}{1000 + 2143} = -54.55.$$

With the load removed and the source zeroed, $R_{\text{out}} = 3000\,\Omega$. The input fraction is 0.6818, so the base signal is 1.364 mV peak and the output amplitude is 109.1 mV peak. R_L affects gain but is removed for R_{out}.

Solution 4 — Early effect with a grounded emitter

First $g_m = 40\,\text{mS}$ and $r_\pi = 2500\,\Omega$. The transistor input path is

$$R_{ib} = r_\pi + (\beta+1)R_E = 2500\,\Omega, \quad R_{\text{in}} = R_B \parallel R_{ib} = 2143\,\Omega.$$

The effective collector load is $R_C \parallel R_L \parallel r_o = 1818\,\Omega$. Hence

$$A_v = -\frac{\beta(1818)}{2500 + (\beta+1)(0)} = -72.73, \qquad A_s = A_v \frac{2143}{1000 + 2143} = -49.59.$$

With the load removed and the source zeroed, $R_{\text{out}} = 2609\,\Omega$. With emitter at AC ground, r_o is another collector-to-ground resistor. It introduces no base-to-collector feedback in this low-frequency hybrid-π model, so R_{in} is unchanged.

Worked solutions: 5–6

Solution 5 — Add emitter degeneration

First $g_m = 40\,\text{mS}$ and $r_\pi = 2500\,\Omega$. The transistor input path is

$$R_{ib} = r_\pi + (\beta+1)R_E = 12600\,\Omega, \quad R_{\text{in}} = R_B \parallel R_{ib} = 6848\,\Omega.$$

The effective collector load is $R_C \parallel R_L = 2000\,\Omega$. Hence

$$A_v = -\frac{\beta(2000)}{2500 + (\beta+1)(100)} = -15.87, \qquad A_s = A_v\frac{6848}{1000+6848} = -13.85.$$

With the load removed and the source zeroed, $R_{\text{out}} = 3000\,\Omega$. The exact relation is $i_e = (\beta+1)i_b$, giving the expressions above. The large-β gain approximation is -16. Degeneration raises input resistance and lowers gain magnitude; with $r_o = \infty$, it does not change $R_{\text{out}} = R_C$.

Solution 6 — Do not silently replace beta plus one

First $g_m = 40\,\text{mS}$ and $r_\pi = 1250\,\Omega$. The transistor input path is

$$R_{ib} = r_\pi + (\beta+1)R_E = 25220\,\Omega, \quad R_{\text{in}} = R_B \parallel R_{ib} = 9406\,\Omega.$$

The effective collector load is $R_C \parallel R_L = 2000\,\Omega$. Hence

$$A_v = -\frac{\beta(2000)}{1250 + (\beta+1)(470)} = -3.965, \qquad A_s = A_v\frac{9406}{1000+9406} = -3.584.$$

With the load removed and the source zeroed, $R_{\text{out}} = 3000\,\Omega$. The exact relation is $i_e = (\beta+1)i_b$, giving the expressions above. The large-β gain approximation is -4.04. Degeneration raises input resistance and lowers gain magnitude; with $r_o = \infty$, it does not change $R_{\text{out}} = R_C$.

Worked solutions: 7–8

Solution 7 — Only part of the emitter resistor is bypassed

For DC, C_E is open and the emitter resistance is $100 + 900 = 1000\,\Omega$. For midband AC, it shorts only R_{E2}, leaving $R_E = 100\,\Omega$. First $g_m = 40\,\text{mS}$ and $r_\pi = 2500\,\Omega$. The transistor input path is

$$R_{ib} = r_\pi + (\beta + 1)R_E = 12600\,\Omega, \quad R_{\text{in}} = R_B \parallel R_{ib} = 7730\,\Omega.$$

The effective collector load is $R_C \parallel R_L = 2000\,\Omega$. Hence

$$A_v = -\frac{\beta(2000)}{2500 + (\beta + 1)(100)} = -15.87, \qquad A_s = A_v \frac{7730}{500 + 7730} = -14.91.$$

With the load removed and the source zeroed, $R_{\text{out}} = 4000\,\Omega$.

Solution 8 — What changes when the bypass capacitor is removed?

First $g_m = 40\,\text{mS}$ and $r_\pi = 2500\,\Omega$. The transistor input path is

$$R_{ib} = r_\pi + (\beta + 1)R_E = 103500\,\Omega, \quad R_{\text{in}} = R_B \parallel R_{ib} = 16760\,\Omega.$$

The effective collector load is $R_C \parallel R_L = 2000\,\Omega$. Hence

$$A_v = -\frac{\beta(2000)}{2500 + (\beta + 1)(1000)} = -1.932, \qquad A_s = A_v \frac{16760}{500 + 16760} = -1.876.$$

With the load removed and the source zeroed, $R_{\text{out}} = 4000\,\Omega$. The DC bias is unchanged because the capacitor was already open for DC. Originally R_{E1} remained unbypassed; after removal both resistors remain in AC. A capacitor bypasses only the component(s) across which it is connected.

Worked solutions: 9–10

Solution 9 — The basic emitter follower

$g_m = 40\,\text{mS}$, $r_\pi = 2500\,\Omega$. The loaded emitter resistance is $R'_E = 500\,\Omega$.

$$R_{ib} = r_\pi + (\beta+1)R'_E = 53000\,\Omega, \quad R_{\text{in}} = R_B \parallel R_{ib} = 53000\,\Omega.$$

$$A_v = \frac{(\beta+1)R'_E}{r_\pi + (\beta+1)R'_E} = 0.9528, \quad A_s = A_v \frac{R_{\text{in}}}{R_s + R_{\text{in}}} = 0.9528.$$

For output resistance, remove R_L and zero v_s. The base now sees $R_s \parallel R_B = 0\,\Omega$:

$$R_{\text{out}} = R_E \parallel \frac{r_\pi + (R_s \parallel R_B)}{\beta+1} = 24.15\,\Omega.$$

The gain is positive and less than unity; the follower supplies impedance transformation.

Solution 10 — Source resistance comes back during Rout

$g_m = 40\,\text{mS}$, $r_\pi = 2500\,\Omega$. The loaded emitter resistance is $R'_E = 500\,\Omega$.

$$R_{ib} = r_\pi + (\beta+1)R'_E = 53000\,\Omega, \quad R_{\text{in}} = R_B \parallel R_{ib} = 14520\,\Omega.$$

$$A_v = \frac{(\beta+1)R'_E}{r_\pi + (\beta+1)R'_E} = 0.9528, \quad A_s = A_v \frac{R_{\text{in}}}{R_s + R_{\text{in}}} = 0.7897.$$

For output resistance, remove R_L and zero v_s. The base now sees $R_s \parallel R_B = 2609\,\Omega$:

$$R_{\text{out}} = R_E \parallel \frac{r_\pi + (R_s \parallel R_B)}{\beta+1} = 48.15\,\Omega.$$

The gain is positive and less than unity; the follower supplies impedance transformation. The $1/g_m$ shortcut ignores base-side source/bias resistance and the parallel internal emitter resistor. It is only an approximation for a strongly driven base and sufficiently large R_E.

Worked solutions: 11–12

Solution 11 — Common base: input at the emitter

$g_m = 40\,\text{mS}$, $r_\pi = 2500\,\Omega$. Looking into the transistor emitter,

$$R_{ie} = \frac{r_\pi + R_B}{\beta + 1} = 24.75\,\Omega, \quad R_{\text{in}} = R_E \parallel R_{ie} = 24.75\,\Omega.$$

Since $v_\pi = -v_i r_\pi/(r_\pi + R_B)$,

$$A_v = +\frac{\beta(R_C \parallel R_L)}{r_\pi + R_B} = 80, \quad A_s = A_v \frac{R_{\text{in}}}{R_s + R_{\text{in}}} = 80.$$

$R_{\text{out}} = R_C = 2000\,\Omega$ because $r_o = \infty$. The stage is noninverting. Raising v_e with fixed base voltage reduces v_{BE} and collector current. The collector resistor then drops less voltage, so v_o rises.

Solution 12 — A high stage gain can hide a small overall gain

$g_m = 40\,\text{mS}$, $r_\pi = 2500\,\Omega$. Looking into the transistor emitter,

$$R_{ie} = \frac{r_\pi + R_B}{\beta + 1} = 24.75\,\Omega, \quad R_{\text{in}} = R_E \parallel R_{ie} = 24.15\,\Omega.$$

Since $v_\pi = -v_i r_\pi/(r_\pi + R_B)$,

$$A_v = +\frac{\beta(R_C \parallel R_L)}{r_\pi + R_B} = 40, \quad A_s = A_v \frac{R_{\text{in}}}{R_s + R_{\text{in}}} = 7.782.$$

$R_{\text{out}} = R_C = 2000\,\Omega$ because $r_o = \infty$. The stage is noninverting. The source resistance is large compared with the roughly $25\,\Omega$ input resistance, so most source voltage is lost before reaching the emitter.

Worked solutions: 13–14

Solution 13 — The base is grounded through a resistor

$g_m = 40\,\text{mS}$, $r_\pi = 2500\,\Omega$. Looking into the transistor emitter,

$$R_{ie} = \frac{r_\pi + R_B}{\beta + 1} = 49.5\,\Omega, \quad R_{\text{in}} = R_E \parallel R_{ie} = 49.5\,\Omega.$$

Since $v_\pi = -v_i r_\pi/(r_\pi + R_B)$,

$$A_v = +\frac{\beta(R_C \parallel R_L)}{r_\pi + R_B} = 20, \quad A_s = A_v \frac{R_{\text{in}}}{R_s + R_{\text{in}}} = 6.623.$$

$R_{\text{out}} = R_C = 2000\,\Omega$ because $r_o = \infty$. The stage is noninverting. $v_\pi/v_i = -1/2$. The finite base resistor carries small-signal base current and therefore develops a nonzero voltage. Only an ideal AC ground fixes $v_b = 0$.

Solution 14 — Compare two base terminations

(i) Grounded base. $g_m = 40\,\text{mS}$, $r_\pi = 2500\,\Omega$. Looking into the transistor emitter,

$$R_{ie} = \frac{r_\pi + R_B}{\beta + 1} = 24.75\,\Omega, \quad R_{\text{in}} = R_E \parallel R_{ie} = 24.15\,\Omega.$$

Since $v_\pi = -v_i r_\pi/(r_\pi + R_B)$,

$$A_v = +\frac{\beta(R_C \parallel R_L)}{r_\pi + R_B} = 40, \quad A_s = A_v \frac{R_{\text{in}}}{R_s + R_{\text{in}}} = 7.782.$$

$R_{\text{out}} = R_C = 2000\,\Omega$ because $r_o = \infty$. The stage is noninverting.

(ii) $R_B = 10\,\text{k}\Omega$**.** $g_m = 40\,\text{mS}$, $r_\pi = 2500\,\Omega$. Looking into the transistor emitter,

$$R_{ie} = \frac{r_\pi + R_B}{\beta + 1} = 123.8\,\Omega, \quad R_{\text{in}} = R_E \parallel R_{ie} = 110.1\,\Omega.$$

Since $v_\pi = -v_i r_\pi/(r_\pi + R_B)$,

$$A_v = +\frac{\beta(R_C \parallel R_L)}{r_\pi + R_B} = 8, \quad A_s = A_v \frac{R_{\text{in}}}{R_s + R_{\text{in}}} = 4.193.$$

$R_{\text{out}} = R_C = 2000\,\Omega$ because $r_o = \infty$. The stage is noninverting. The source divider improves, but the base moves with the emitter, reducing $|v_\pi/v_i|$. Overall gain falls; input resistance alone cannot predict gain.

Worked solutions: 15–16

Solution 15 — CE driving a follower

Stage 1 has $g_{m1} = 40\,\text{mS}$, $r_{\pi 1} = 2.5\,\text{k}\Omega$, and $R_{\text{in1}} = 20\,\text{k}\Omega \parallel 2.5\,\text{k}\Omega = 2222\,\Omega$. For stage 2, $g_{m2} = 80\,\text{mS}$, $r_{\pi 2} = 1.25\,\text{k}\Omega$, and $R'_E = 1\,\text{k}\Omega \parallel R_L = 333.3\,\Omega$.

$$R_{\text{in2}} = 50\,\text{k}\Omega \parallel [1250 + 101R'_E] = 20560\,\Omega.$$

Stage 2 loads stage 1. Therefore

$$A_1 = \frac{v_x}{v_i} = -g_{m1}(3\,\text{k}\Omega \parallel R_{\text{in2}}) = -104.7, \quad A_2 = \frac{v_o}{v_x} = \frac{101R'_E}{1250 + 101R'_E} = 0.9642.$$

$$A_v = A_1 A_2 = -101, \qquad A_s = A_v \frac{R_{\text{in1}}}{1000 + R_{\text{in1}}} = -69.63.$$

For the final output resistance, remove R_L. With $v_s = 0$, the follower base sees $R_{C1} \parallel R_{B2} = 3\,\text{k}\Omega \parallel 50\,\text{k}\Omega$. Thus

$$R_{\text{out}} = 1\,\text{k}\Omega \parallel \frac{1250 + (3\,\text{k}\Omega \parallel 50\,\text{k}\Omega)}{101} = 38.83\,\Omega.$$

Do not use the loaded R_{in2} as the source resistance at its own base.

Solution 16 — Change the final load and follow the effect backward

Stage 1 has $g_{m1} = 40\,\text{mS}$, $r_{\pi 1} = 2.5\,\text{k}\Omega$, and $R_{\text{in1}} = 20\,\text{k}\Omega \parallel 2.5\,\text{k}\Omega = 2222\,\Omega$. For stage 2, $g_{m2} = 80\,\text{mS}$, $r_{\pi 2} = 1.25\,\text{k}\Omega$, and $R'_E = 1\,\text{k}\Omega \parallel R_L = 47.62\,\Omega$.

$$R_{\text{in2}} = 50\,\text{k}\Omega \parallel [1250 + 101R'_E] = 5405\,\Omega.$$

Stage 2 loads stage 1. Therefore

$$A_1 = \frac{v_x}{v_i} = -g_{m1}(3\,\text{k}\Omega \parallel R_{\text{in2}}) = -77.17, \quad A_2 = \frac{v_o}{v_x} = \frac{101R'_E}{1250 + 101R'_E} = 0.7937.$$

$$A_v = A_1 A_2 = -61.25, \qquad A_s = A_v \frac{R_{\text{in1}}}{1000 + R_{\text{in1}}} = -42.24.$$

For the final output resistance, remove R_L. With $v_s = 0$, the follower base sees $R_{C1} \parallel R_{B2} = 3\,\text{k}\Omega \parallel 50\,\text{k}\Omega$. Thus

$$R_{\text{out}} = 1\,\text{k}\Omega \parallel \frac{1250 + (3\,\text{k}\Omega \parallel 50\,\text{k}\Omega)}{101} = 38.83\,\Omega.$$

Do not use the loaded R_{in2} as the source resistance at its own base. Lower R_L lowers the follower's input resistance, which loads the CE collector more heavily. Output resistance is unchanged here because the external load is removed during that measurement and the bias currents are held fixed.

Worked solutions: 17–18

Solution 17 — Two CE stages: do not multiply unloaded gains

Both transistors have $g_m = 40\,\text{mS}$ and $r_\pi = 2.5\,\text{k}\Omega$.

$$R_{\text{in1}} = 20\,\text{k}\Omega \parallel 2.5\,\text{k}\Omega = 2222\,\Omega, \qquad R_{\text{in2}} = 10\,\text{k}\Omega \parallel 2.5\,\text{k}\Omega = 2\,\text{k}\Omega.$$

The first collector sees $R_{C1} \parallel (R_x + R_{\text{in2}})$, with $R_x = 0\,\Omega$:

$$\frac{v_x}{v_i} = -.04[3000 \parallel (0 + 2000)] = -48.$$

$$\frac{v_{b2}}{v_x} = \frac{2000}{0 + 2000} = 1, \quad \frac{v_o}{v_{b2}} = -.04(2000 \parallel 2000) = -40.$$

$$A_v = 1920, \qquad A_s = A_v \frac{2222}{500 + 2222} = 1567.$$

The two inversions produce a positive total gain. $R_{\text{out}} = R_{C2} = 2\,\text{k}\Omega$ after removal of R_L. The unloaded-gain product would be $(-120)(-80) = 9600$, while the actual base-input-to-final-output gain is 1920. Both the interstage load and final load must be included.

Solution 18 — An extra interstage resistor

Both transistors have $g_m = 40\,\text{mS}$ and $r_\pi = 2.5\,\text{k}\Omega$.

$$R_{\text{in1}} = 20\,\text{k}\Omega \parallel 2.5\,\text{k}\Omega = 2222\,\Omega, \qquad R_{\text{in2}} = 10\,\text{k}\Omega \parallel 2.5\,\text{k}\Omega = 2\,\text{k}\Omega.$$

The first collector sees $R_{C1} \parallel (R_x + R_{\text{in2}})$, with $R_x = 1000\,\Omega$:

$$\frac{v_x}{v_i} = -.04[3000 \parallel (1000 + 2000)] = -60.$$

$$\frac{v_{b2}}{v_x} = \frac{2000}{1000 + 2000} = 0.6667, \quad \frac{v_o}{v_{b2}} = -.04(2000 \parallel 2000) = -40.$$

$$A_v = 1600, \qquad A_s = A_v \frac{2222}{500 + 2222} = 1306.$$

The two inversions produce a positive total gain. $R_{\text{out}} = R_{C2} = 2\,\text{k}\Omega$ after removal of R_L. The first collector is less heavily loaded, so its local gain magnitude rises. But R_x introduces a second divider; the overall gain falls. Trace the voltage through every intermediate node.

Worked solutions: 19–20

Solution 19 — From the DC circuit to the AC circuit

The DC Thevenin source is $V_{th} = 8(20/80) = 2$ V and $R_{th} = 60\,\text{k}\Omega \parallel 20\,\text{k}\Omega = 15\,\text{k}\Omega$.

$$I_B = \frac{2 - .7}{15000 + 101(1000)} = 11.21\,\mu\text{A}, \quad I_C = 1.121\,\text{mA}.$$

$$V_E = 1.132\,\text{V}, \quad V_B = 1.832\,\text{V}, \quad V_C = 4.638\,\text{V}.$$

$V_{CE} = 3.506$ V and $V_C > V_B$, so forward-active operation is consistent. For midband, the emitter is grounded by C_E, the supply is AC ground, and the bias resistors become $15\,\text{k}\Omega$ from base to ground. Draw r_π from base to emitter and a controlled current $g_m v_\pi$ from collector to emitter. First $g_m = 44.83\,\text{mS}$ and $r_\pi = 2231\,\Omega$. The transistor input path is

$$R_{ib} = r_\pi + (\beta + 1)R_E = 2231\,\Omega, \quad R_{\text{in}} = R_B \parallel R_{ib} = 1942\,\Omega.$$

The effective collector load is $R_C \parallel R_L = 2000\,\Omega$. Hence

$$A_v = -\frac{\beta(2000)}{2231 + (\beta + 1)(0)} = -89.66, \qquad A_s = A_v \frac{1942}{0 + 1942} = -89.66.$$

With the load removed and the source zeroed, $R_{\text{out}} = 3000\,\Omega$.

Solution 20 — Remove the bypass and add a realistic source

First $g_m = 44.83\,\text{mS}$ and $r_\pi = 2231\,\Omega$. The transistor input path is

$$R_{ib} = r_\pi + (\beta + 1)R_E = 103200\,\Omega, \quad R_{\text{in}} = R_B \parallel R_{ib} = 13100\,\Omega.$$

The effective collector load is $R_C \parallel R_L = 2000\,\Omega$. Hence

$$A_v = -\frac{\beta(2000)}{2231 + (\beta + 1)(1000)} = -1.937, \qquad A_s = A_v \frac{13100}{1000 + 13100} = -1.8.$$

With the load removed and the source zeroed, $R_{\text{out}} = 3000\,\Omega$. The capacitor was already open in DC, and C_{in} isolates the source resistance from the bias network. Thus the bias current and small-signal parameters remain those of Exercise 19.

Worked solutions: 21–22

Solution 21 — Challenge: Rout with degeneration and finite ro

$g_m = .04$ S, $r_\pi = 2500\,\Omega$, $r_o = 50000\,\Omega$. The zeroed ideal source grounds the base, so let $E = R_E \parallel r_\pi = 185.2\,\Omega$. Apply v_x to the collector. Since $v_\pi = -v_e$, emitter KCL gives

$$\frac{v_e}{E} + \frac{v_e - v_x}{r_o} + g_m v_e = 0, \quad v_e = \frac{v_x}{1 + r_o/E + g_m r_o}.$$

The test current into the transistor is $i_x = (v_x - v_e)/r_o - g_m v_e$. Substitution gives

$$R_{\text{out,Q}} = r_o + (1 + g_m r_o)(R_E \parallel r_\pi) = 420600\,\Omega.$$

$$R_{\text{out}} = R_C \parallel R_{\text{out,Q}} = 3962\,\Omega.$$

The shortcut $4\,\text{k}\Omega \parallel 50\,\text{k}\Omega = 3704\,\Omega$ ignores the emitter motion and the feedback created by degeneration. The idealized $r_o = \infty$ answer would be $4\,\text{k}\Omega$.

Solution 22 — Challenge: finite ro in a follower

$g_m = 40\,\text{mS}$, $r_\pi = 2500\,\Omega$. The loaded emitter resistance is $R'_E = 400\,\Omega$.

$$R_{ib} = r_\pi + (\beta + 1)R'_E = 42900\,\Omega, \quad R_{\text{in}} = R_B \parallel R_{ib} = 13640\,\Omega.$$

$$A_v = \frac{(\beta + 1)R'_E}{r_\pi + (\beta + 1)R'_E} = 0.9417, \quad A_s = A_v \frac{R_{\text{in}}}{R_s + R_{\text{in}}} = 0.7719.$$

For output resistance, remove R_L and zero v_s. The base now sees $R_s \parallel R_B = 2609\,\Omega$:

$$R_{\text{out}} = R_E \parallel r_o \parallel \frac{r_\pi + (R_s \parallel R_B)}{\beta + 1} = 47.01\,\Omega.$$

The gain is positive and less than unity; the follower supplies impedance transformation. Because the collector is AC ground, the collector–emitter resistor r_o is an emitter-to-ground resistor. Include it in the loaded emitter resistance for gain, and retain it after the external load is removed for output resistance.

Worked solutions: 23–24

Solution 23 — Design for a specified loaded gain

$g_m = .04$ S, $r_\pi = 2500\,\Omega$, and $R_C \parallel R_L = 2000\,\Omega$. Use the exact finite-$\beta$ expression:

$$-10 = -\frac{100(2000)}{2500 + 101R_E} \quad \Rightarrow \quad R_E = \frac{20000 - 2500}{101} = 173.3\,\Omega.$$

$$R_{\text{in}} = 50\,\text{k}\Omega \parallel 20\,\text{k}\Omega = 14290\,\Omega, \quad A_s = -10\frac{14290}{1000 + 14290} = -9.346, \quad R_{\text{out}} = 4\,\text{k}\Omega.$$

The source divider still reduces v_i relative to v_s. The question fixes a loaded stage gain, not a source-to-output gain.

Solution 24 — Diagnose a complete wrong solution

First $g_m = 40\,\text{mS}$ and $r_\pi = 2500\,\Omega$. The transistor input path is

$$R_{ib} = r_\pi + (\beta + 1)R_E = 12600\,\Omega, \quad R_{\text{in}} = R_B \parallel R_{ib} = 6848\,\Omega.$$

The effective collector load is $R_C \parallel R_L = 2000\,\Omega$. Hence

$$A_v = -\frac{\beta(2000)}{2500 + (\beta + 1)(100)} = -15.87, \qquad A_s = A_v\frac{6848}{1000 + 6848} = -13.85.$$

With the load removed and the source zeroed, $R_{\text{out}} = 3000\,\Omega$. Mistakes: (1) R_s is outside the amplifier input port, R_B is omitted, and emitter resistance is not reflected to the base. (2) Gain omits both the collector load and degeneration. (3) Source attenuation is omitted. (4) The external load must be removed for output resistance. The corrected output peak amplitude is 13.85 mV for a 1 mV source.